Anton R. Banta, Ph.D.

Electrical and Biotechnology Engineer · Machine Learning / Data Analysis Expert

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PROFESSIONAL SUMMARY

Machine Learning and Data Science Engineer with a Ph.D. in Electrical and Computer Engineering and cross-disciplinary expertise in computational neuroscience, signal processing, and applied AI. Experienced in developing and deploying machine learning models for large complex datasets, ranging from neural and physiological signals to multimodal sensor and imaging data. Skilled at translating research concepts into scalable analytical tools, predictive models, and technology prototypes. Passionate about applying data-driven innovation to improve outcomes across industries including healthcare, biotechnology, human performance, and advanced sensing.

AREAS OF EXPERTISE

Technical Skills: Python (NumPy, Pandas, SciPy, Scikit-learn, PyTorch, TensorFlow, Keras, Jupyter) | MATLAB | SQL /PostgreSQL |AWS (S3, EC2, SageMaker) | Docker | Git / GitHub | Data Visualization (Matplotlib, Seaborn, Plotly) | Sensors & IoT Systems

Professional Skills: Cross-functional Collaboration | Technical Communication | Research Translation | Experimental Design | Agile / Iterative Development | Problem Solving | Adaptability | Innovation

PROFESSIONAL EXPERIENCE

ŌURA, San Diego, *Senior Algorithm Scientist* *January 2026 - Present*

- Lead cross-disciplinary initiatives spanning hardware, firmware, data science, and product to translate novel ring sensing modalities into user-facing health features.
- Design and execute human subject studies to evaluate signal quality, robustness, and real-world performance of next-generation optical, mechanical, and physiological sensors.
- Implement advanced signal processing and machine learning algorithms in Python (NumPy, SciPy, PyTorch) to extract robust physiological biomarkers from multimodal time-series data.

Houston Methodist Hospital, Houston, *Biotech Engineer - Machine Learning* *January 2024 – January 2026*

- Designed and deployed **machine learning models** to predict cerebrospinal fluid (CSF) flow from multimodal biosignals (EEG, ECG, PPG, respiration), using advanced **feature engineering** and **regularized regression** to drive clinical insight.
- Built and trained **LSTM networks** to decode motor, sensory, and autonomic activity from spinal cord signals, supporting next-generation **brain–computer interface (BCI)** development.
- Analyzed **large neural datasets** from freely moving primates to study continual learning across behavioral contexts.
- Leveraged **foundation models** and **diffusion architectures** to reconstruct visual perception from invasive neural recordings.

Rice University, Houston, *Research Assistant* *August 2019 – December 2023*

- Developed **novel signal processing and machine learning algorithms** for the analysis of physiological data for cardiac and neurology applications
- Worked independently and in **large multidisciplinary teams** to solve medical data problems, resulting in **four publications**

Sensorum, New York City, *AI PhD Summer Intern* *May 2022 - August 2022*

- Created a **lightweight audio event detection model** for cough detection on embedded devices, optimized for **low-latency edge deployment**.
- Improved model training efficiency by 3× through custom **learning rate scheduling** and model tuning.
- Delivered a **signal preprocessing pipeline** that doubled detection sensitivity at equivalent false-positive rates, fully integrated via company's **CI/CD pipeline**.

EDUCATION

Rice University, Houston, TX, *Ph.D. candidate in Electrical and Computer Engineering* *August 2019 – December 2023*

- GPA: 3.81 / 4.00
- Relevant Coursework — Neural Signal Processing, Introduction to Deep Neural Networks, Introduction to Random Processes, Embedded Machine Learning, Network Science, Introduction to Machine Learning, Nano-Neurotechnology, Theoretical Neuroscience, Human-Robot Interaction, Optimization, Quantum Information Science and Technology.

PAPERS AND PATENTS

Banta AR, Fedorov D, Altilia A, Malliaras GG, Barone DG. Brain–computer interfaces and neuroprosthetics in the next era of neurosurgery. British Journal of Neurosurgery. 2026 Jun 14:1-9.

Banta A, D'Ambrosio R, Karmonik C, Mathur S, Golanov EV, Regnier-Golanov A, Aazhang B, Britz GW. Quantification of Perivascular Flow Dynamics of Human Cerebrospinal Fluid Along Major Arteries Using Phase-Contrast MRI. Radiology. 2025 Sep 23;316(3):e243521.

El Hadwe S, Ruiz-Mateos Serrano R, Psaltakis G, Forner M, Lee C, Swedick S, **Banta A**, Zhang X, Ghannam M, Hasan T, Carnicer-Lombarte A. Surface circumferential spinal cord recording in freely moving rodents. Nature Communications. 2026 Jul 13.

Banta A, et al. Methods and devices for the non-invasive assessment of cerebro-spinal dynamics and diagnosis of brain pathophysiologies. Patent No. WO2024211033A2 issued 2024.

Banta A, Cosentino R, John MM, Post A, Buchan S, Razavi M, Aazhang B. A novel convolutional neural network for reconstructing surface electrocardiograms from intracardiac electrograms and vice versa. Artificial intelligence in medicine. 2021 Aug 1;118:102135.